Implementation Plan BTP

📁 PROJECT STRUCTURE

```
asha-copilot/
│
├── backend/
│   ├── main.py                          # FastAPI entry point (Adithya — Part 4)
│   │
│   ├── agents/
│   │   ├── __init__.py
│   │   ├── supervisor.py                # LangGraph supervisor (Adithya — Part 4)
│   │   ├── structuring_agent.py         # Speech → JSON (Rohan — Part 1)
│   │   ├── protocol_check_agent.py      # RAG + red flags (Anoushka — Part 3)
│   │   ├── scheme_match_agent.py        # PMMVY/JSY/JSSK rules (Rohan — Part 1)
│   │   ├── report_generator.py          # Report + tasks (Rohan — Part 2)
│   │   └── response_composer.py         # Merge outputs (Rohan — Part 1)
│   │
│   ├── asr/
│   │   ├── __init__.py
│   │   ├── transcriber.py               # IndicWhisper/Conformer (Rohan — Part 1)
│   │   └── preprocessor.py              # Audio preprocessing (Rohan — Part 1)
│   │
│   ├── tts/
│   │   ├── __init__.py
│   │   └── synthesizer.py               # Bhashini/Coqui TTS (Rohan — Part 1)
│   │
│   ├── rag/
│   │   ├── __init__.py
│   │   ├── embeddings.py                # Sentence transformers (Anoushka — Part 3)
│   │   └── retriever.py                 # ChromaDB retrieval (Anoushka — Part 3)
│   │
│   ├── rules/
│   │   ├── __init__.py
│   │   ├── red_flags.py                 # Hardcoded rules (Anoushka — Part 3)
│   │   └── schemes.py                   # PMMVY/JSY/JSSK rules (Rohan — Part 1)
│   │
│   ├── db/
│   │   ├── __init__.py
│   │   ├── models.py                    # SQLModel models (Anoushka — Part 3)
│   │   └── database.py                  # SQLite/PostgreSQL (Adithya — Part 4)
│   │
│   └── schemas/
│       ├── __init__.py
│       └── medical_record.py            # Pydantic schemas (Anoushka — Part 3)
│
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   ├── MicInput.jsx             # Mic recording (Rohan — Part 2)
│   │   │   ├── VisitHistory.jsx         # Past visits (Rohan — Part 2)
│   │   │   ├── TaskList.jsx             # Follow-up tasks (Rohan — Part 2)
│   │   │   └── ReportView.jsx           # Report display (Rohan — Part 2)
│   │   ├── App.jsx                      # Main app (Rohan — Part 2)
│   │   └── main.jsx                     # Entry point (Rohan — Part 2)
│   └── package.json                     # Dependencies (Rohan — Part 2)
│
```

```
├── data/
│   ├── guidelines/                    # NHM/ICMR PDFs (Anoushka — Part 3)
│   ├── schemes/                       # Scheme rules (Rohan — Part 1)
│   └── evaluation/                    # Test set (Rohan — Part 1)
│
├── evaluation/
│   ├── metrics.py                     # WER, F1, etc. (Adithya — Part 4)
│   ├── ablation.py                    # Multi vs single agent (Adithya — Part 4)
│   └── results/                       # Evaluation outputs (Adithya — Part 4)
│
├── docs/
│   ├── architecture.md                # System design (Adithya — Part 4)
│   ├── setup.md                       # Setup guide (Adithya — Part 4)
│   └── evaluation.md                  # Evaluation plan (Adithya — Part 4)
│
├── requirements.txt                   # Dependencies (Adithya — Part 4)
├── docker-compose.yml                 # Docker setup (Adithya — Part 4)
└── README.md                          # Documentation (Adithya — Part 4)
```

S.No.	File	Owner	Part
1	`backend/asr/transcriber.py`	**Rohan**	Part 1
2	`backend/asr/preprocessor.py`	**Rohan**	Part 1
3	`backend/tts/synthesizer.py`	**Rohan**	Part 1
4	`backend/agents/structuring_agent.py`	**Rohan**	Part 1
5	`backend/agents/scheme_match_agent.py`	**Rohan**	Part 1
6	`backend/agents/response_composer.py`	**Rohan**	Part 1
7	`backend/rules/schemes.py`	**Rohan**	Part 1
8	`data/evaluation/`	**Rohan**	Part 1
9	`data/schemes/`	**Rohan**	Part 1
10	`backend/agents/report_generator.py`	**Rohan**	Part 2
11	`frontend/src/components/MicInput.jsx`	**Rohan**	Part 2
12	`frontend/src/components/VisitHistory.jsx`	**Rohan**	Part 2
13	`frontend/src/components/TaskList.jsx`	**Rohan**	Part 2
14	`frontend/src/components/ReportView.jsx`	**Rohan**	Part 2
15	`frontend/src/App.jsx`	**Rohan**	Part 2
16	`frontend/src/main.jsx`	**Rohan**	Part 2
17	`frontend/package.json`	**Rohan**	Part 2
18	`backend/agents/protocol_check_agent.py`	**Anoushka**	Part 3
19	`backend/rag/embeddings.py`	**Anoushka**	Part 3
20	`backend/rag/retriever.py`	**Anoushka**	Part 3
21	`backend/rules/red_flags.py`	**Anoushka**	Part 3
22	`backend/schemas/medical_record.py`	**Anoushka**	Part 3

S.No.	File	Owner	Part
23	`backend/db/models.py`	**Anoushka**	Part 3
24	`data/guidelines/`	**Anoushka**	Part 3
25	`backend/agents/supervisor.py`	**Adithya**	Part 4
26	`backend/main.py`	**Adithya**	Part 4
27	`backend/db/database.py`	**Adithya**	Part 4
28	`evaluation/metrics.py`	**Adithya**	Part 4
29	`evaluation/ablation.py`	**Adithya**	Part 4
30	`evaluation/results/`	**Adithya**	Part 4
31	`docs/architecture.md`	**Adithya**	Part 4
32	`docs/setup.md`	**Adithya**	Part 4
33	`docs/evaluation.md`	**Adithya**	Part 4
34	`requirements.txt`	**Adithya**	Part 4
35	`docker-compose.yml`	**Adithya**	Part 4
36	`README.md`	**Adithya**	Part 4

Rohan — Part 1 + Part 2

Part	Files
Part 1	transcriber, preprocessor, synthesizer, structuring_agent, scheme_match_agent, response_composer, schemes, data/evaluation, data/schemes
Part 2	report_generator, MicInput, VisitHistory, TaskList, ReportView, App, main, package.json

Anoushka — Part 3

Part	Files
Part 3	protocol_check_agent, embeddings, retriever, red_flags, medical_record, models, data/guidelines

Adithya (C) — Part 4

Part	Files
Part 4	supervisor, main, database, metrics, ablation, results, architecture, setup, evaluation, requirements, docker-compose, README

📅 PHASE 1: SETUP & FOUNDATION (September)

Environment ready, basic pipeline working

Rohan (Part 1 + Part 2)

Build	File
Install Python, torchaudio, librosa, transformers, FastAPI, React	`requirements.txt`
Create project folder structure	All folders
Build ASR transcribe function	`backend/asr/transcriber.py`
Build audio preprocessor	`backend/asr/preprocessor.py`
Build TTS synthesize function	`backend/tts/synthesizer.py`
Build Pydantic schema for medical record	`backend/schemas/medical_record.py`
Build Structuring Agent (basic extraction)	`backend/agents/structuring_agent.py`
Build Scheme Match Agent	`backend/agents/scheme_match_agent.py`
Build scheme eligibility rules	`backend/rules/schemes.py`
Build Response Composer	`backend/agents/response_composer.py`

Anoushka (Part 3)

Build	File
Download NHM guidelines PDFs	`data/guidelines/`
Download ICMR guidelines PDFs	`data/guidelines/`
Build embeddings function	`backend/rag/embeddings.py`
Build retriever (chunk + store in ChromaDB)	`backend/rag/retriever.py`
Build red-flag rules	`backend/rules/red_flags.py`
Build Pydantic schema	`backend/schemas/medical_record.py`
Build DB models	`backend/db/models.py`

Adithya (Part 4)

Build	File
Set up FastAPI app	`backend/main.py`
Set up SQLite connection	`backend/db/database.py`
Build /transcribe endpoint	`backend/main.py`
Build /structure endpoint	`backend/main.py`

Build	File
Build /protocol-check endpoint	`backend/main.py`
Build /scheme-match endpoint	`backend/main.py`
Build /report-generate endpoint	`backend/main.py`
Build /full-pipeline endpoint	`backend/main.py`

Deliverables (End of September)

Person	Working Deliverable
Rohan	ASR transcribes audio → text, TTS converts text → audio, Scheme Match Agent working
Anoushka	ChromaDB has NHM/ICMR chunks, retriever returns relevant passages
Adithya	All API endpoints respond with JSON

PHASE 2: CORE AGENTS (October)

Goal: All 4 agents working individually

Rohan (Part 1 + Part 2)

Build	File
Build Structuring Agent (few-shot extraction)	`backend/agents/structuring_agent.py`
Test with sample transcripts	Notes
Build Report Generator Agent	`backend/agents/report_generator.py`
Test with sample structured records	Notes
Build Response Composer (merge outputs)	`backend/agents/response_composer.py`
Test merge logic	Notes
Build Frontend MicInput component	`frontend/src/components/MicInput.jsx`
Build Frontend VisitHistory component	`frontend/src/components/VisitHistory.jsx`

Anoushka (Part 3)

Build	File
Build Protocol Check Agent (RAG + red flags)	`backend/agents/protocol_check_agent.py`
Test with sample structured records	Notes
Add citation generation to Protocol Agent	`backend/agents/protocol_check_agent.py`
Add more red-flag rules	`backend/rules/red_flags.py`
Test all rules with edge cases	Notes

Adithya (Part 4)

Build	File
Build LangGraph supervisor	`backend/agents/supervisor.py`
Test routing logic	Notes
Integrate supervisor with API	`backend/main.py`
Test full pipeline	Notes
Build evaluation metrics (WER, F1)	`evaluation/metrics.py`
Build ablation study script	`evaluation/ablation.py`
Test metrics with sample data	Notes
Build evaluation results folder	`evaluation/results/`

Deliverables (End of October)

Person	Working Deliverable
Rohan	Structuring + Scheme Match + Report Generator + Response Composer + 2 Frontend components
Anoushka	Protocol Check working with citations
Adithya	Supervisor routes correctly, Metrics + Ablation ready

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📅 PHASE 3: INTEGRATION (November)

All agents connected end-to-end

Rohan (Part 1 + Part 2)

Build	File
Build Frontend TaskList component	`frontend/src/components/TaskList.jsx`
Build Frontend ReportView component	`frontend/src/components/ReportView.jsx`
Build Frontend App layout	`frontend/src/App.jsx`
Build Frontend entry point	`frontend/src/main.jsx`
Build Frontend package.json	`frontend/package.json`
Connect Frontend to API	Integration
Test full voice pipeline	Integration

Anoushka (Part 3)

Build	File
Refine Protocol Check with citations	`backend/agents/protocol_check_agent.py`

Build	File
Test red-flag recall	Notes
Refine RAG retrieval quality	`backend/rag/retriever.py`
Add more guidelines to ChromaDB	`data/guidelines/`
Test all agents together	Integration

Adithya (Part 4)

Build	File
Connect supervisor to all agents	`backend/agents/supervisor.py`
Test agent routing	Notes
Add persistence layer	`backend/db/database.py`
Save visits to DB	Notes
Build end-to-end pipeline	`backend/main.py`
Test full pipeline	Integration
Fix integration bugs	All files

Deliverables (End of November)

Person	Working Deliverable
Rohan	Complete Frontend + Voice pipeline integrated
Anoushka	Protocol Agent tested
Adithya	End-to-end pipeline working

📅 PHASE 4: TESTING & REFINEMENT (December)

System works end-to-end with voice confirmation

Rohan (Part 1 + Part 2)

Build	File
Test ASR → Structuring → Protocol → Scheme → Report	Integration
Test TTS read-back	Integration
Test voice confirmation loop	Integration
Fix ASR errors	Notes
Test with noisy audio	Notes
Test Frontend UI	Integration

Build	File
Finalize voice pipeline	All files

Anoushka (Part 3)

Build	File
Test Protocol Agent with clinical vignettes	Notes
Measure red-flag recall	Notes
Add more red-flag rules	`backend/rules/red_flags.py`
Test edge cases	Notes
Finalize Protocol Agent	All files

Adithya (Part 4)

Build	File
Test API with all agents	Integration
Add error handling	`backend/main.py`
Test persistence layer	`backend/db/database.py`
Test data flow	Notes
Test end-to-end with voice	Integration
Fix integration bugs	All files
Finalize integration	All files

Deliverables (End of December)

Person	Working Deliverable
Rohan	Complete voice-to-voice pipeline working
Anoushka	Protocol Agent finalized
Adithya	Full end-to-end pipeline with persistence

📅 PHASE 5: EVALUATION SETUP (January)

Evaluation data ready, system stable

Rohan (Part 1 + Part 2)

Build	File
Record 50-100 simulated ASHA visits	`data/evaluation/`
Transcribe and annotate	`data/evaluation/`

Build	File
Test ASR on evaluation set	Notes
Measure WER	Notes
Test Frontend with real data	Integration
Fix Frontend bugs	Notes
Finalize evaluation set	`data/evaluation/`

Anoushka (Part 3)

Build	File
Test Protocol on evaluation set	Notes
Measure agent accuracy	Notes
Refine RAG on evaluation set	`backend/rag/retriever.py`
Test red-flag recall	Notes
Finalize Protocol evaluation	All files

Adithya (Part 4)

Build	File
Build evaluation scripts	`evaluation/metrics.py`
Set up metrics calculation	`evaluation/metrics.py`
Run evaluation	`evaluation/results/`
Collect metrics	`evaluation/results/`
Build ablation study	`evaluation/ablation.py`
Run multi-agent vs single-agent	Notes
Finalize evaluation	All files

Deliverables (End of January)

Person	Working Deliverable
Rohan	Evaluation set (50-100 samples) with WER + Frontend stable
Anoushka	Protocol evaluation done
Adithya	Complete evaluation results + ablation study

 PHASE 6: DOCUMENTATION (February)

All docs ready, system polished

Rohan (Part 1 + Part 2)

Build	File
Write ASR documentation	Docs
Write Frontend documentation	Docs
Create ASR plots	`evaluation/results/`
Create Frontend screenshots	`evaluation/results/`
Write Phase 6 report section	Report
Prepare presentation slides	Slides
Finalize all documentation	Docs

Anoushka (Part 3)

Build	File
Write Protocol documentation	Docs
Write RAG documentation	Docs
Create Protocol accuracy plots	`evaluation/results/`
Create RAG retrieval plots	`evaluation/results/`
Write Phase 6 report section	Report
Prepare presentation slides	Slides
Finalize all documentation	Docs

Adithya (Part 4)

Build	File
Write architecture documentation	`docs/architecture.md`
Write setup documentation	`docs/setup.md`
Write evaluation documentation	`docs/evaluation.md`
Write README	`README.md`
Write Phase 6 report section	Report
Prepare presentation slides	Slides
Finalize all documentation	Docs

Deliverables (End of February)

Person	Working Deliverable
Rohan	ASR + Frontend documentation + plots
Anoushka	Protocol + RAG documentation + plots
Adithya	Architecture + Setup + Evaluation docs + README

📅 PHASE 7: FINAL EVALUATION & DEMO (March)

Final report, demo ready

Rohan (Part 1 + Part 2)

Build	File
Run final ASR evaluation	`evaluation/results/`
Document WER results	Report
Write ASR evaluation section	Report
Create final ASR plots	`evaluation/results/`
Write final report sections	Report
Prepare demo	Demo
Practice presentation	Slides

Anoushka (Part 3)

Build	File
Run final Protocol evaluation	`evaluation/results/`
Document agent accuracy	Report
Write Protocol evaluation section	Report
Create final Protocol plots	`evaluation/results/`
Write final report sections	Report
Prepare demo	Demo
Practice presentation	Slides

Adithya (Part 4)

Build	File
Run final ablation study	`evaluation/results/`
Document comparison	Report
Write ablation section	Report
Create final comparison plots	`evaluation/results/`
Write final report sections	Report
Prepare demo	Demo
Practice presentation	Slides

Deliverables (End of March)

Person	Working Deliverable
Rohan	ASR evaluation report + Frontend demo
Anoushka	Protocol evaluation report + RAG demo
Adithya	Ablation study report + complete final documentation + demo

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